

Aryan Karawale

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Education

• Bharati Vidyapeeth College of Engineering, Katraj, Pune

2023-27

BTech Computer Engineering CGPA: 8.21

Projects

• AI-Powered Movie Recommender System | Python, TensorFlow, SQLite

A personalized recommendation engine using Deep Learning to map user-item interactions in a latent embedding space.

- Engineered a Neural Collaborative Filtering (NCF) model using TensorFlow/Keras and embedding layers to predict user preferences with high precision.
- Developed an Explainable AI (XAI) module using Cosine Similarity to provide real-time transparency into the model's recommendation logic.
- Designed an automated cold-start strategy that transitions from popularity-based metrics to personalized AI rankings as user interaction data increases.
- Technology Used: Python, TensorFlow, Keras, Scikit-learn, Pandas, SQLite3, Streamlit, YAML, BCrypt.

• SecureMe: Personal Security App | Kotlin, Jetpack Compose, Android SDK

A rapid-response Android application built with Kotlin and Jetpack Compose for personal safety.

- Engineered an SOS system using FusedLocationProviderClient and SmsManager to instantly dispatch automated SMS alerts with live GPS tracking links.
- Implemented a background shake-to-alert trigger via Android's SensorManager, enabling hands-free emergency activation even when the device is locked.
- Developed active deterrence features (panic sirens, simulated calls) and built a modern offline-first UI using Jetpack Compose and SharedPreferences.
- Built a modern, declarative user interface using Jetpack Compose and Material Design 3, featuring local data persistence (SharedPreferences) for secure, offline-first emergency contact management.

• Responsive Developer Portfolio Website | React.js, Tailwind CSS, Vercel

A portfolio website where you can see my other works and connect with me.

- Architecture: Developed a high-performance, single-page application (SPA) using React.js, utilizing a component-based architecture for maximum code reusability and maintainability.
- UI/UX Design: Engineered a modern, fully responsive user interface with Tailwind CSS, incorporating a mobilefirst design approach to ensure seamless cross-device compatibility.
- Interactivity: Implemented advanced UI animations and transitions using Framer Motion, significantly enhancing user engagement and visual storytelling.

Certifications

- Computer Graphics: scored 70% (NPTEL)
- Data Base Management System: scored 60% (NPTEL)

Technical Skills and Interests

- **Languages/ Frameworks:** C/C++, Python, Javascript, Node.js, React, Express.js & FastAPI
- **Software Tools:** Git, Blender
- **Cloud/Databases:** MongoDB, Relational Database(mysql)
- **Areas of Interest:** Full Stack Web Development, Computer Graphics

Extra- Curricular Participation

• LEO Member, LEO BVUCOEP

Jul 2024 - Apr 2025

- Supported various social causes and activities as part of team LEO.

• Associate Member, Google Developers Groups on Campus(GDGC), BVUCOEP

Aug 2024 - Apr 2025

- Enhanced my Web-Dev skills through various events organised by the Cell.